

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 20%

QUESTIONS: 5

**Time Duration: 1 Hour
Total Marks:30**

Annexure A

Resolution Algorithm Implementation

Code of Conduct:

I Hemalakshmi H - 192571049 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that **the ground is wet** using the Resolution Algorithm.

Tasks

1. Represent the statements in propositional logic.
2. Convert all statements into CNF.

Negate the goal.

1. Apply the Resolution Algorithm step by step.
2. Show the Empty Clause ($\square$) if the goal is proved.

Question 2 – Student Assignment Submission

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks
2. Rahul submitted the assignment.

Goal

Prove that **Rahul receives internal marks** using the Resolution Algorithm.

Tasks

1. Write the propositional logic expressions.
2. Convert the premises into CNF.

Negate the goal.

1. Apply the Resolution Algorithm.
2. Conclude whether the goal is proved.

Question 3 – Library Membership

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

1. Prove that **Priya can borrow books** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base using propositional logic.
2. Convert each statement into CNF.

Negate the goal.

1. Perform resolution until no new clauses can be generated.
2. State the final conclusion.

Question 4 – Placement Eligibility

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

1. Prove that **Arun is eligible for placement** using the Resolution Algorithm.

Tasks

1. Convert the statements into propositional logic.
2. Write the CNF clauses.
3. Negate the goal.
4. Apply the Resolution Algorithm step by step.
5. Draw the resolution tree and conclude.

Question 5 – Access Control System

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that **the user is granted access** using the Resolution Algorithm.

Tasks

1. Represent the knowledge base in propositional logic.
2. Convert all implications into CNF.
3. Negate the goal.
4. Apply the Resolution Algorithm to derive the Empty Clause ($\square$).
5. Explain each resolution step and write the final conclusion.

1.

RUBRICS FOR CALCULATION

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Question 1:

1. Represent the statements in propositional logic

Let:

- R : "It rains"
- W : "The ground is wet"

Knowledge Base (KB):

1. $R \rightarrow W$
2. R

Goal:

- Prove W

2. Convert all statements into CNF

P1: $R \rightarrow W$

- Eliminate implication: $\neg R \vee W$
- CNF: $(\neg R \vee W)$

P2: R

- Already atomic
- CNF: (R)

Goal: W

- For resolution refutation, add negation of goal: $\neg W$

So the clause set is:

- C1: $\neg R \vee W$
- C2: R
- C3: $\neg W$ (negated goal)

3. Apply the Resolution Algorithm step by step

Step 1: Resolve C1 and C2

- C1: $\neg R \vee W$
- C2: R

Resolution on $R / \neg R$:

- Resulting clause: C4: W

Step 2: Resolve C4 and C3

- C4: W
- C3: $\neg W$

Resolution on $W / \neg W$:

- Resulting clause: C5: $\square$ (empty clause)

4. Conclusion

Since we derived the empty clause ($\square$) from $KB \wedge \neg \text{Goal}$, the goal W ("the ground is wet") is logically entailed by the knowledge base using resolution.

Question 2:

1. Propositional logic expressions

Let:

- S : "Student submits the assignment"

- M : “Student receives internal marks”
- Specifically for Rahul:
 - S_R : “Rahul submitted the assignment”
 - M_R : “Rahul receives internal marks”

Knowledge Base (KB):

1. If a student submits the assignment, then the student receives internal marks:
 $S \rightarrow M$

2. Rahul submitted the assignment:

$$S_R$$

Goal:

- Prove: M_R

We treat “Rahul” as a specific instance of the general rule, so S_R implies M_R .

2. Convert premises into CNF and negate the goal

P1: $S \rightarrow M$

- Eliminate implication: $\neg S \vee M$
- CNF: $(\neg S \vee M)$

P2: S_R

- Atomic, CNF: (S_R)

To apply resolution, we instantiate the general rule for Rahul:

- From $S \rightarrow M$, for Rahul: $S_R \rightarrow M_R$
- CNF: $(\neg S_R \vee M_R)$

So our CNF clauses are:

- C1: $\neg S_R \vee M_R$
- C2: S_R

Negate the goal:

Goal: M_R Negated goal: $\neg M_R$

- C3: $\neg M_R$

3. Apply the Resolution Algorithm

Step 1: Resolve C1 and C2

- C1: $\neg S_R \vee M_R$
- C2: S_R

Resolution on $S_R / \neg S_R$:

- Result: C4: M_R

Step 2: Resolve C4 and C3

- C4: M_R
- C3: $\neg M_R$

Resolution on $M_R / \neg M_R$:

- Result: C5: $\square$ (empty clause)

4. Conclusion

Since the empty clause ($\square$) is derived from $KB \wedge \neg\text{Goal}$, the goal M_R (“Rahul receives internal marks”) is **proved** by the resolution algorithm.

Question 3:

1. Represent the knowledge base in propositional logic

Let:

- L : “Person is a library member”
- B : “Person can borrow books” For Priya specifically:
- L_P : “Priya is a library member”
- B_P : “Priya can borrow books”

Knowledge Base (KB):

1. If a person is a library member, then the person can borrow books:

$$L \rightarrow B$$

2. Priya is a library member:

$$L_P$$

Goal:

- Prove: B_P

We use the rule instantiated for Priya: $L_P \rightarrow B_P$.

2. Convert each statement into CNF and negate the goal

Rule for Priya:

$$L_P \rightarrow B_P$$

- Eliminate implication: $\neg L_P \vee B_P$
- CNF clause: C1: $(\neg L_P \vee B_P)$

Fact:

$$L_P$$

- CNF clause: C2: (L_P)

Goal:

$$B_P$$

Negate the goal for resolution refutation:

- $\neg B_P$
- CNF clause: C3: $(\neg B_P)$

Clause set:

- C1: $\neg L_P \vee B_P$
- C2: L_P
- C3: $\neg B_P$

3. Perform resolution

Step 1: Resolve C1 and C2

- C1: $\neg L_P \vee B_P$
- C2: L_P

Resolution on $L_P / \neg L_P$:

- Result: C4: (B_P)

Step 2: Resolve C4 and C3

- C4: B_P
- C3: $\neg B_P$

Resolution on $B_P / \neg B_P$:

- Result: C5: $\square$ (empty clause)

No further clauses can be generated.

4. Final conclusion:

Since the empty clause ($\square$) is derived from $KB \wedge \neg \text{Goal}$, the goal B_P ("Priya can borrow books") is logically proved using the Resolution Algorithm.

Question 4:

1. Convert the statements into propositional logic

Let:

- A : "Student clears the aptitude test"
- E : "Student is eligible for placement"

For Arun:

- A_A : "Arun cleared the aptitude test"
- E_A : "Arun is eligible for placement"

Knowledge Base (KB):

1. If a student clears the aptitude test, then the student is eligible for placement:

$$A \rightarrow E$$

For Arun: $A_A \rightarrow E_A$

2. Arun cleared the aptitude test:

$$A_A$$

Goal:

$$E_A$$

2. Write the CNF clauses

Rule for Arun:

$$A_A \rightarrow E_A$$

- Eliminate implication: $\neg A_A \vee E_A$
- CNF clause: C1: $(\neg A_A \vee E_A)$

Fact:

$$A_A$$

- CNF clause: C2: (A_A)

Goal:

$$E_A$$

Negated goal for refutation:

- $\neg E_A$
- CNF clause: C3: $(\neg E_A)$

Clause set:

- C1: $\neg A_A \vee E_A$
- C2: A_A
- C3: $\neg E_A$

3. Apply the Resolution Algorithm step by step

Step 1: Resolve C1 and C2

- C1: $\neg A_A \vee E_A$
- C2: A_A

Resolution on $A_A / \neg A_A$:

- Result: C4: (E_A)

Step 2: Resolve C4 and C3

- C4: E_A
- C3: $\neg E_A$

Resolution on $E_A / \neg E_A$:

- Result: C5: $\square$ (empty clause)

4. Resolution tree (textual)

- $\square$
 - from: E_A (C4) and $\neg E_A$ (C3)
 - E_A (C4) from: $\neg A_A \vee E_A$ (C1) and A_A (C2)

So the tree is:

- Root: $\square$
 - Child: E_A
 - Parents: $\neg A_A \vee E_A, A_A$
 - Child: $\neg E_A$

5. Conclusion

Since the empty clause ($\square$) is derived from $KB \wedge \neg \text{Goal}$, the goal E_A ("Arun is eligible for placement") is proved by the Resolution Algorithm.

Question 5:

1. Represent the knowledge base in propositional logic

Let:

- P : "User enters the correct password"
- A : "User is authenticated"
- G : "User is granted access"

Knowledge Base (KB):

1. If a user enters the correct password, then the user is authenticated:
$$P \rightarrow A$$
2. If a user is authenticated, then the user is granted access:
$$A \rightarrow G$$
3. The user entered the correct password:
$$P$$

Goal:

$$G$$

2. Convert all implications into CNF

P1: $P \rightarrow A$

- Eliminate implication: $\neg P \vee A$
- CNF clause: C1: $(\neg P \vee A)$

P2: $A \rightarrow G$

- Eliminate implication: $\neg A \vee G$
- CNF clause: C2: $(\neg A \vee G)$

P3: P

- Atomic, CNF clause: C3: (P)

3. Negate the goal

Goal: G Negated goal: $\neg G$

- CNF clause: C4: $(\neg G)$

Clause set:

- C1: $\neg P \vee A$
- C2: $\neg A \vee G$
- C3: P
- C4: $\neg G$

4. Apply the Resolution Algorithm

Step 1: From correct password to authentication

Resolve C1 and C3:

- C1: $\neg P \vee A$
- C3: P

Resolution on $P / \neg P$:

- Result: C5: (A)

So we derive: **User is authenticated.**

Step 2: From authentication to granted access

Resolve C2 and C5:

- C2: $\neg A \vee G$
- C5: A

Resolution on $A / \neg A$:

- Result: C6: (G)

So we derive: **User is granted access.**

Step 3: Resolve with negated goal

Resolve C6 and C4:

- C6: G

- C4: $\neg G$

Resolution on $G/\neg G$:

- Result: C7: $\square$ (empty clause)

5. Final conclusion

Since the empty clause ($\square$) is derived from $KB \wedge \neg \text{Goal}$, the goal G (“the user is granted access”) is logically entailed by the knowledge base using the Resolution Algorithm.